

**Q:** What is middleware in Express?

**A:** Functions that execute during the request-response cycle and can modify req/res or end the cycle.

**Q:** Middleware lifecycle?

**A:** Request → Middleware chain → Route handler → Response → Error middleware (if triggered).

**Q:** Application-level middleware?

**A:** Bound to app using app.use(), runs for all routes.

**Q:** Router-level middleware?

**A:** Attached to express.Router(), scoped to specific routes.

**Q:** Error-handling middleware?

**A:** Takes (err, req, res, next), handles runtime errors.

**Q:** Third-party middleware?

**A:** External packages like body-parser, cors, morgan.

**Q:** Request flow in Express?

**A:** Client → Middleware → Router → Controller → Response.

**Q:** Blocking vs Non-blocking?

**A:** Blocking waits for task completion; non-blocking uses async callbacks/promises.

**Q:** Body-parser?

**A:** Parses incoming request body into JSON or URL-encoded format.

**Q:** SSR vs CSR?

**A:** SSR renders HTML on server; CSR renders in browser using JS.

**Q:** Template engines?

**A:** Tools like EJS/HBS to generate dynamic HTML on server.

**Q:** EJS usage?

**A:** Set view engine to ejs, use <%= %> to embed JS.

**Q:** HBS usage?

**A:** Handlebars syntax {{}} for templating.

**Q:** SQL vs NoSQL?

**A:** SQL is relational (tables); NoSQL is flexible (documents, key-value).

**Q:** MongoDB connection?

**A:** Use `mongoose.connect(uri)` to connect.

**Q:** ODM Mongoose?

**A:** Maps MongoDB data to JS objects with schemas and models.

**Q:** Cookies?

**A:** Small data stored in browser.

**Q:** Sessions?

**A:** Server-side storage of user data linked via session ID.

**Q:** Express-session?

**A:** Middleware for managing sessions.

**Q:** Bcrypt?

**A:** Hashes passwords securely.

**Q:** JWT?

**A:** Token-based authentication with `header.payload.signature`.

**Q:** Passport.js?

**A:** Authentication middleware supporting multiple strategies.

**Q:** Socket.io?

**A:** Enables real-time, full-duplex communication between client and server.

**Q:** Full duplex?

**A:** Two-way communication simultaneously.

**Q:** PostgreSQL?

**A:** Open-source relational database with strong ACID compliance.